



CS 540 Introduction to Artificial Intelligence

Linear Algebra & PCA

University of Wisconsin-Madison
Spring 2025

Announcements

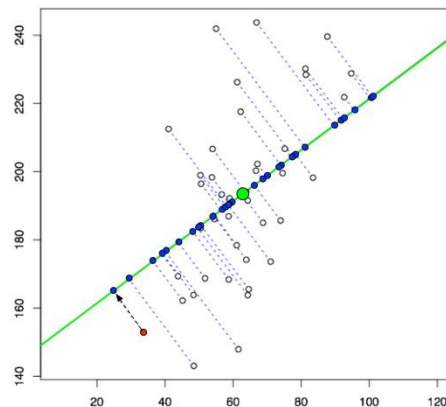
- **HW 1 online:**
 - Writing assignment---nothing too stressful
 - Deadline **Tuesday Feb. 4th 11:59PM**
- **HW 2:**
 - Probability
 - Deadline **Thursday Feb. 6th 11:59PM**

- Class roadmap:

Linear Algebra and PCA	} Mostly Foundations
Logic	
NLP	
Machine Learning: Introduction	
Machine Learning: Unsupervised Learning I	

Outline

- Basics: vectors, matrices, operations
- Dimensionality reduction
- Principal Components Analysis (PCA)



Lior Pachter

Matrix Inverses

- If for A there is a B such that $AB = BA = I$
 - Then A is invertible/nonsingular, B is its inverse
 - Some matrices are **not** invertible!
 - Usual notation: A^{-1}

$\rightarrow = A A^{-1}$

$\rightarrow$ this is the Identity Matrix

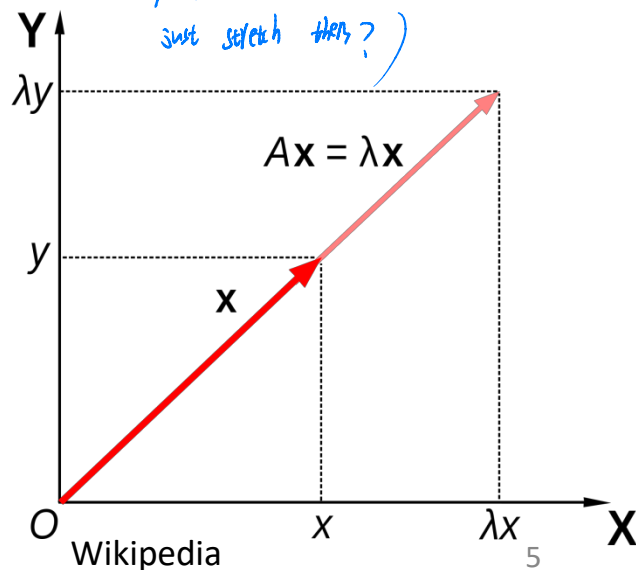
$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \times \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix} = I$$

Eigenvalues & Eigenvectors

• For a square matrix A , solutions to $Av = \lambda v$

- v (nonzero) is a vector: **eigenvector**
- λ is a scalar: **eigenvalue**

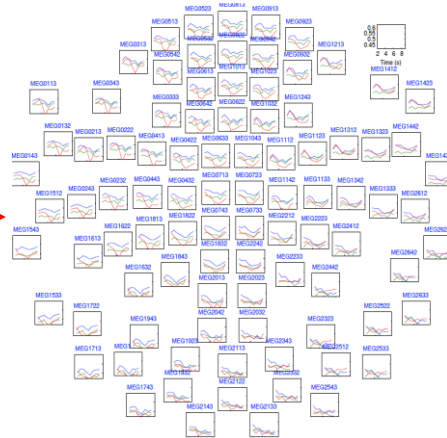
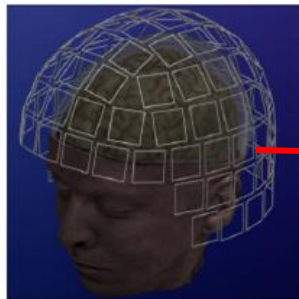
- Intuition: A is a linear transformation;
- Can stretch/rotate vectors;
- E-vectors: only stretched (by e-vals)



Dimensionality Reduction

- Vectors store features. Lots of features!
 - Document classification: thousands of words per doc
 - Netflix surveys: 480189 users x 17770 movies
 - **MEG Brain Imaging**: 120 locations x 500 time points x 20 objects

	movie 1	movie 2	movie 3
Tom	5	?	?
George	?	?	3
Susan	4	3	1
Beth	4	3	?



Dimensionality Reduction

Reduce dimensions

- Why?
 - Lots of features redundant
 - Storage & computation costs



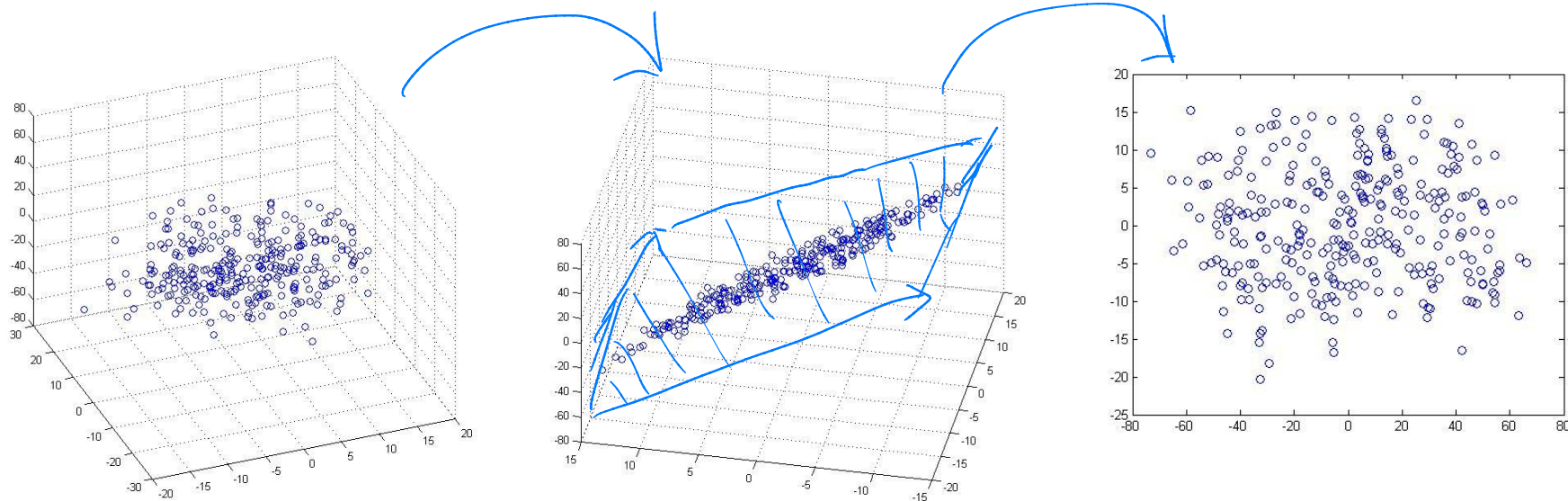
CreativeBlop

- ★ • Goal: take $x \in \mathbb{R}^d \rightarrow x \in \mathbb{R}^r$ for $r \ll d$
 - But, minimize information loss

Dimensionality Reduction

Examples: 3D to 2D

From 3D to 2D, we reduce 1-D!



Andrew Ng

Break & Quiz

Q 2.1: What is the inverse of $A = \begin{bmatrix} 0 & 2 \\ 3 & 0 \end{bmatrix}$

A: $A^{-1} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix}$

B: $A^{-1} = \begin{bmatrix} 0 & \frac{1}{3} \\ \frac{1}{2} & 0 \end{bmatrix}$

C: Undefined / A is not invertible

Break & Quiz

Q 2.1: What is the inverse of $A = \begin{bmatrix} 0 & 2 \\ 3 & 0 \end{bmatrix}$

A: $A^{-1} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix}$

$$AA^{-1} = \begin{bmatrix} 0 & 2 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 * a + 2 * c & 0 * b + 2 * d \\ 3 * a + 0 * c & 3 * b + 0 * d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

B: $A^{-1} = \begin{bmatrix} 0 & \frac{1}{3} \\ \frac{1}{2} & 0 \end{bmatrix}$

$$2c = 1$$

$$3a = 0$$

$$2d = 0$$

$$3b = 1$$

C: Undefined / A is not invertible

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & 1/3 \\ 1/2 & 0 \end{bmatrix}$$

Break & Quiz

Q 2.2: What are the eigenvalues of $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- A. -1, 2, 4
- B. 0.5, 0.2, 1.0
- C. 0, 2, 5
- D. 2, 5, 1

Break & Quiz

Q 2.2: What are the eigenvalues of $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- A. -1, 2, 4
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- D. 2, 5, 1**

Solution #1: You may recall from a linear algebra course that the eigenvalues of a diagonal matrix (in which only diagonal entries are non-zero) are just the entries along the diagonal. Hence D is the correct answer.

Break & Quiz

Q 2.2: What are the eigenvalues of $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Solution #2: Use the definition of eigenvectors and values: $Av = \lambda v$

- A. -1, 2, 4
- B. 0.5, 0.2, 1.0
- C. 0, 2, 5
- D. 2, 5, 1**

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 2v_1 + 0v_2 + 0v_3 \\ 0v_1 + 5v_2 + 0v_3 \\ 0v_1 + 0v_2 + 1v_3 \end{bmatrix} = \begin{bmatrix} 2v_1 \\ 5v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} \lambda v_1 \\ \lambda v_2 \\ \lambda v_3 \end{bmatrix}$$

Since A is a 3×3 matrix, A has 3 eigenvalues and so there are 3 combinations of values for λ and v that will satisfy the above equation.

The simple form of the equations suggests starting by checking each of the standard basis vectors* as v and then solving for λ . Doing so gives D as the correct answer.

*Each standard basis vector $e_i \in \mathbb{R}^n$ is the vector in which all components are zero except component i is 1.

Break & Quiz

Q 2.3: Suppose we are given a dataset with $n=10000$ samples with 100-dimensional binary feature vectors. Our storage device has a capacity of 50000 bits. What's the lowest compression ratio we can use?

- A. 20X
- B. 100X
- C. 5X
- D. 1X

Break & Quiz

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Break & Quiz

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A. 20X

B. 100X

C. 5X

D. 1X

50,000 bits / 10,000 samples
means compressed version must
have 5 bits / sample.

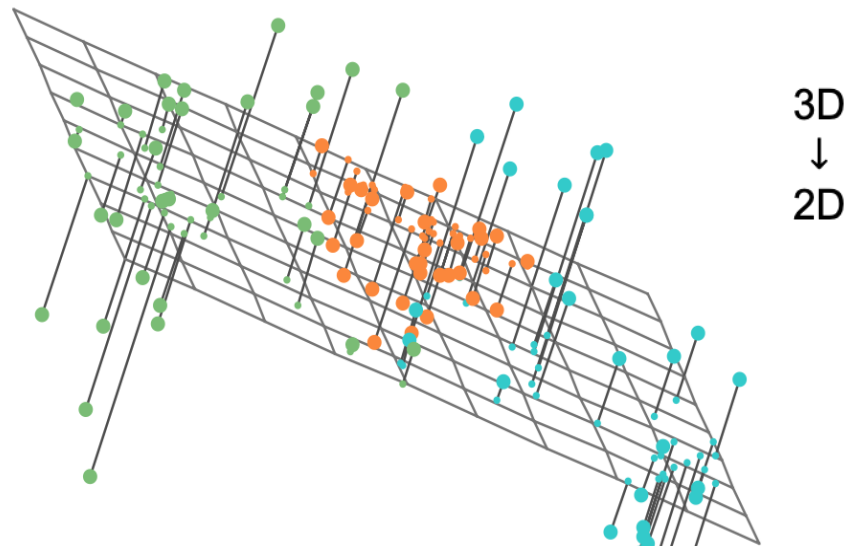
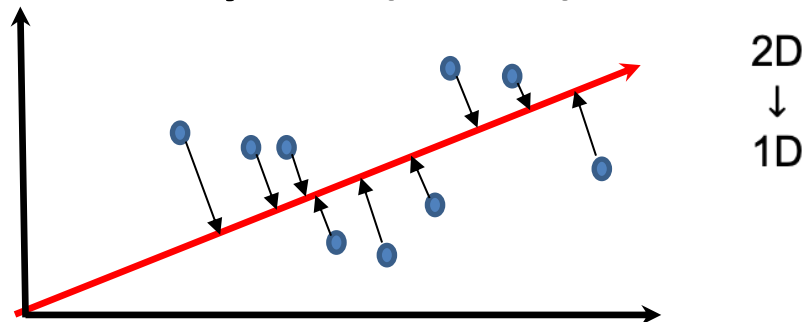
Dataset has 100 bits / sample.

Must compress 20x smaller to fit on
device.

Principal Components Analysis (PCA)

- A type of dimensionality reduction approach

- ★ • For when data is approximately lower dimensional

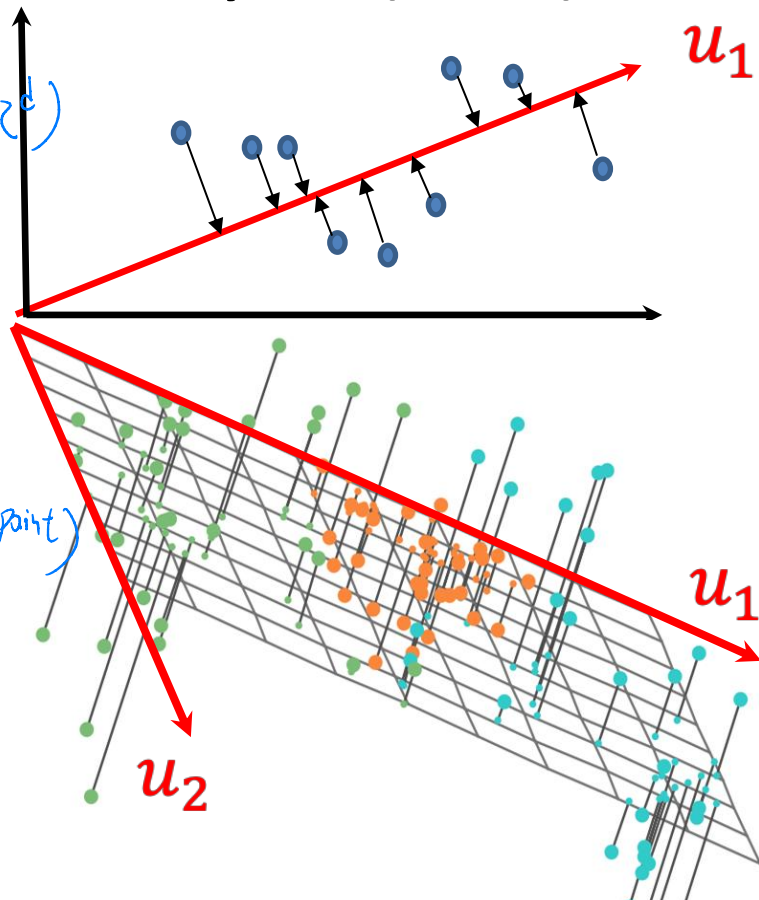


Principal Components Analysis (PCA)

- Find axes $u_1, u_2, \dots, u_m \in \mathbb{R}^d$ of a subspace *(we initially have $x_1, \dots, x_n \in \mathbb{R}^d$)*
 - Will project to this subspace

- ★* Want to preserve data
 - minimize projection error
★ (Minimize distance between x and new data point)

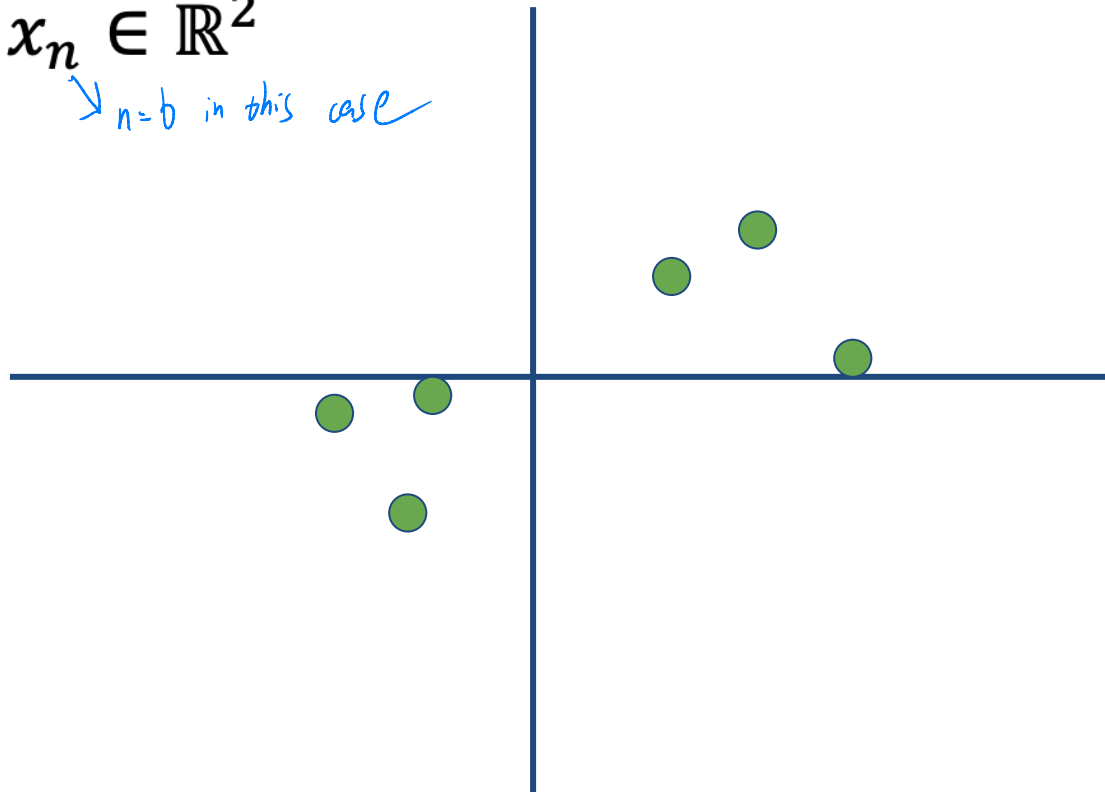
- ★* These vectors are the principal components



Projection: An Example

$$x_1, x_2, \dots, x_n \in \mathbb{R}^2$$

$\rightarrow n=6$ in this case

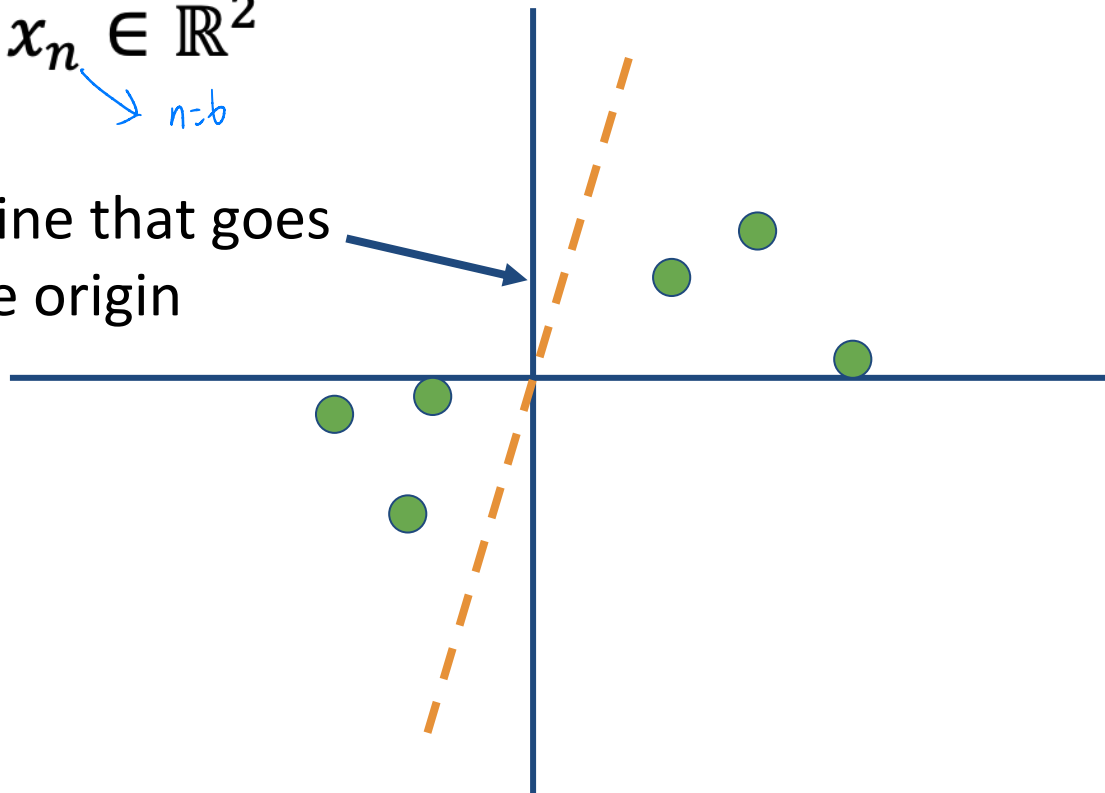


Projection: An Example

$$x_1, x_2, \dots, x_n \in \mathbb{R}^2$$

n=6

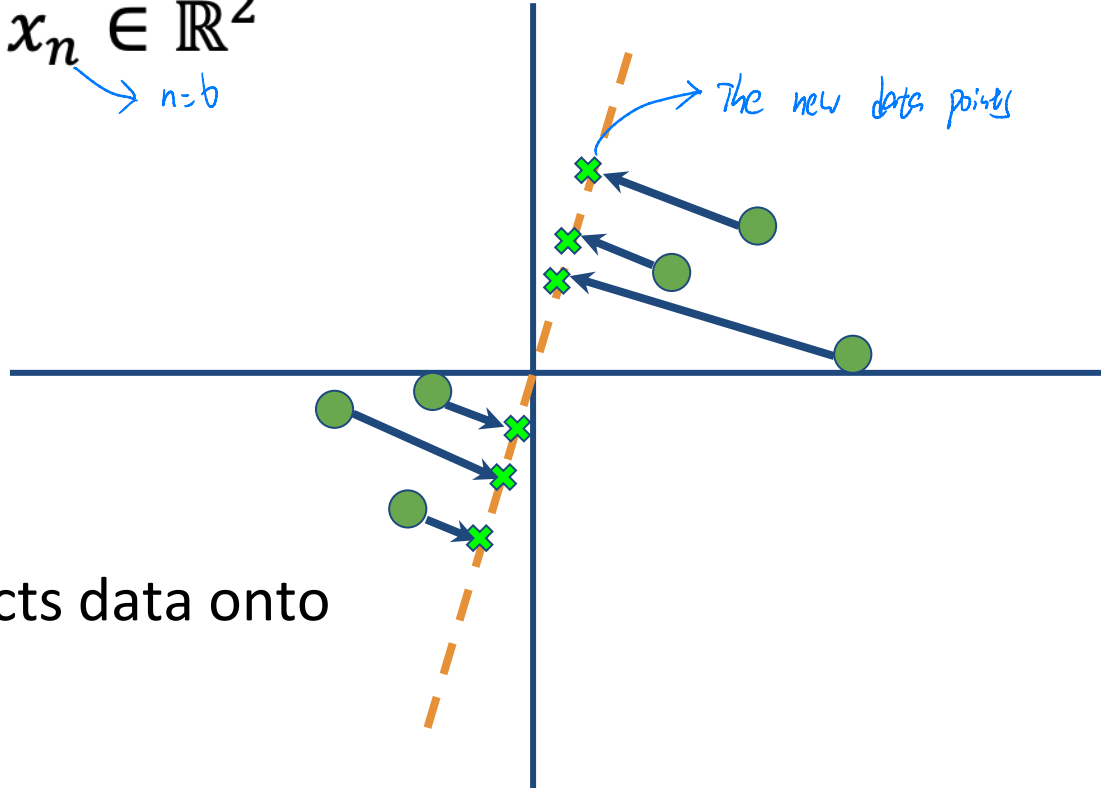
A random line that goes through the origin



Projection: An Example

$$x_1, x_2, \dots, x_n \in \mathbb{R}^2$$

$\rightarrow n=6$

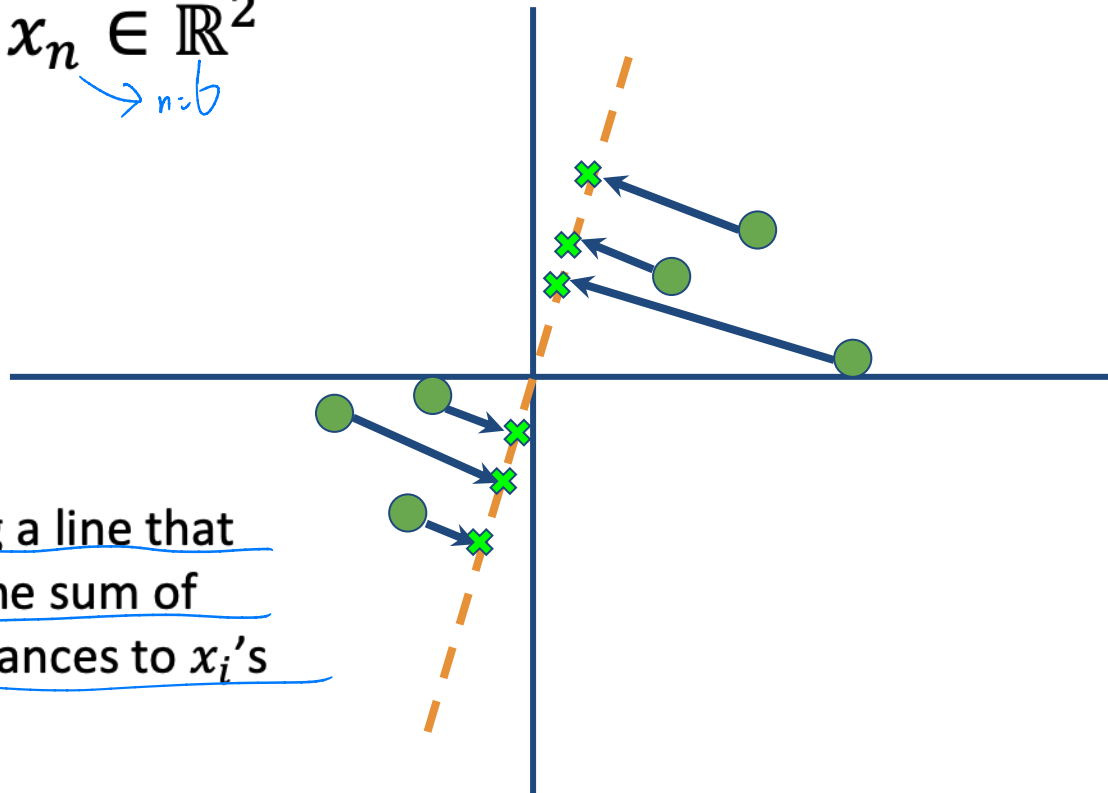


PCA projects data onto
this line

Projection: An Example

$$x_1, x_2, \dots, x_n \in \mathbb{R}^2$$

n=6

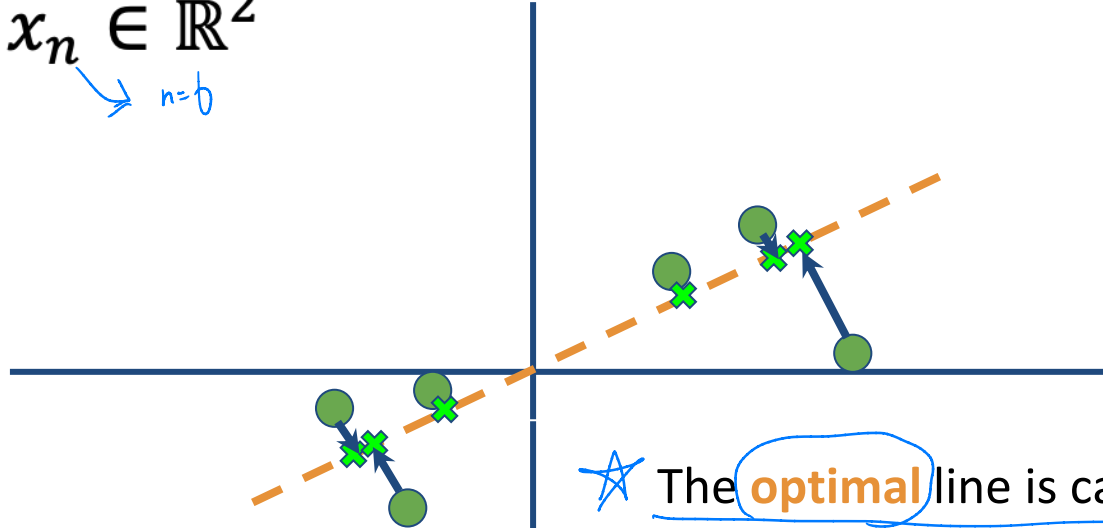


★ Goal: finding a line that
minimizes the sum of
squared distances to x_i 's

Projection: An Example

$$x_1, x_2, \dots, x_n \in \mathbb{R}^2$$

n=6



★ The optimal line is called Principal Component 1

The sum of squared distances gets smaller as the line fits better

→ then repeat this process for next best principal component (that should be orthogonal to u_1)



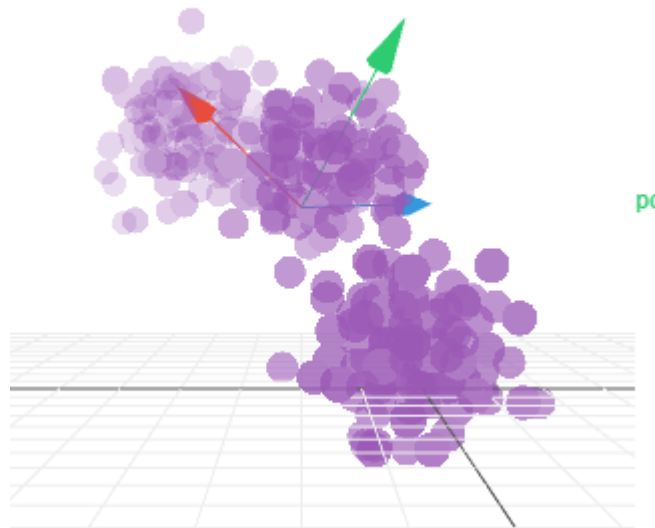
PCA Procedure

Inputs: data $x_1, x_2, \dots, x_n \in \mathbb{R}^d$



Center data so that $\frac{1}{n} \sum_{i=1}^n x_i = 0$

(Idea: the axes may not only pass through the origin, thus we want to shift the data such that the average is 0)



Victor Powell

PCA Procedure

Output:

principal components $u_1, \dots, u_m \in \mathbb{R}^d$

- Orthogonal

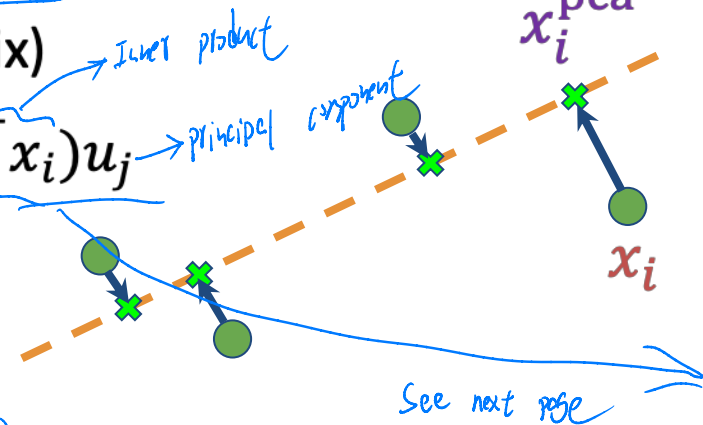
- Can show: they are top- m **eigenvectors** of

$$S = \frac{1}{n-1} \sum_{i=1}^n x_i x_i^T \text{ (covariance matrix)}$$

★ – Each x_i projected to $x_i^{\text{pca}} = \sum_{j=1}^m (u_j^T x_i) u_j$

★ (S is $d \times d$ matrix, we don't need to iteratively find each top principal component, just use this formula for one function call from Python [shortcut here])

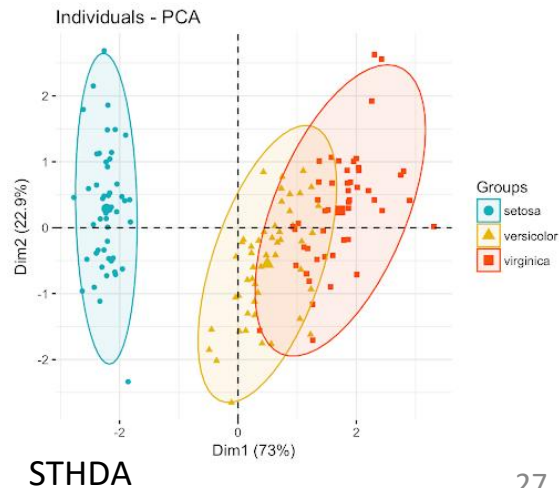
(Indicates the corresponding top m max eigenvalues)



★ (For X_i^{PCA} , there are still n of the points, each point is selected with n eigenvalues)

Many Variations

- PCA, Kernel PCA, ICA, CCA
 - Extract structure from high dimensional dataset
- Uses:
 - **Visualization** : One advantage of PCA
 - Efficiency
 - Noise removal
 - Downstream machine learning use



Application: Image Compression

- Start with image; divide into 12x12 patches

$$x \in \mathbb{R}^{144}$$

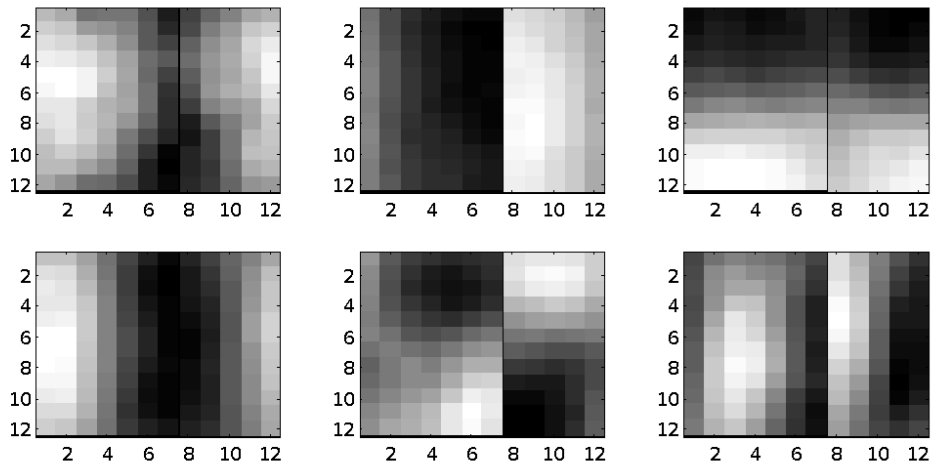
- That is, 144-D vector

- **Original image:**



Application: Image Compression

- 6 principal components (as an image)



Application: Image Compression

- Project to 6D



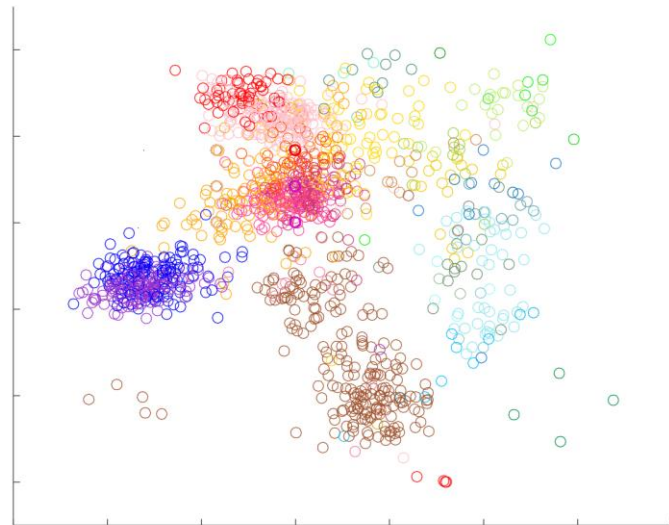
Compressed



Original

Application: Exploratory Data Analysis

- [Novembre et al. '08]: Take top two singular vectors of people x SNP matrix (POPRES)



“Genes Mirror Geography in Europe”

Readings

- Vast literature on linear algebra.
- Local class: **Math 341**
- More on PCA (and other matrix methods in ML): **CS 532**
- **Suggested reading:**
 - Lecture notes on PCA by Roughgarden and Valiant
<https://web.stanford.edu/class/cs168/l/l7.pdf>
 - 760 notes by Zhu <https://pages.cs.wisc.edu/~jerryzhu/cs760/PCA.pdf>